Chapter 14

Complex Amalgam Restorations

"Simplicities are enormously complex".

—Richard O. Moore

Complex posterior restorations are used to replace any missing structure of teeth that have fractured, have severe caries involvement or have existing restorative material. These restorations usually involve the replacement of one or more missing cusps and require additional means of retention. This chapter describes the use of dental amalgam for complex direct posterior restorations.

Indications

Complex posterior restorations are indicated when tooth structure is missing due to cusp fracture, severe caries lesion development or when replacement of existing restorative material is necessary. Complex amalgam preparations should be considered when large amounts of tooth structure are missing, when one or more cusps need to be covered and when increased resistance and retention forms are needed (Fig. 14.1).[1]

Complex amalgams may be used as follows:

I. **Definitive (final) restorations**

Usually a weakened tooth is best restored with a properly designed indirect (usually cast) restoration that prevents tooth fracture caused by mastication forces (Chapter 23). In selected cases, amalgam preparations that improve the resistance form of a tooth may be designed (Fig. 14.2).

When conventional retention features are not adequate because of insufficient remaining tooth structure, the retention form may be enhanced by using auxiliary features such as *slots* and *pins*. The type of retention features needed depends on the amount of tooth structure remaining and the tooth being restored. As more tooth structure is lost, more auxiliary retention is required. Slots and pins provide additional resistance and retention form to the tooth when remaining vertical walls are inadequate.

II. **Foundation restoration**

Although complex amalgam restorations are used occasionally as an alternative to indirect restorations, particularly due to cost savings, they are often used as foundations for full-coverage restorations. Abutment teeth for fixed prostheses may use a complex restoration as a foundation (Fig. 14.3).

III. **Control restorations in teeth that have a questionable pulpal or periodontal prognosis**

A tooth with a severe caries lesion that may require endodontic therapy or crown lengthening or that has an uncertain periodontal prognosis is often treated initially with a *control restoration*. A control restoration helps

i. protect the pulp from the oral cavity (i.e. fluids, thermal stresses, pH changes, bacteria);
ii. provide an anatomic contour that is consistent with gingival health;
iii. facilitate control of acidogenic biofilm and resultant caries risk; and
iv. provide some resistance against tooth fracture (or propagation of an existing fracture). (See Chapter 2 for caries-control rationale and techniques.)

The status and prognosis of the tooth determine the size, number and placement of retention features. Larger restorations generally require more retention. The size, number and location of retention features



Figure 14.1 Mesioocclusodistolingual (MODL) complex amalgam in a maxillary first molar.

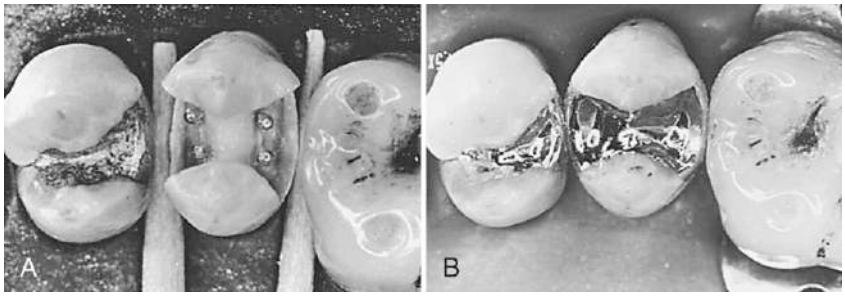


Figure 14.2 Maxillary second premolar weakened by an extensive caries lesion and by the small fracture line extending mesiodistally on the centre of the excavated dentinal wall. (A) Minikin pins placed in the gingival floor improve resistance form after amalgam has been placed. (B) Restorations polished.



Figure 14.3 Mesioocclusodistofacial (MODF) amalgam foundation for abutment tooth (maxillary molar) in preparation for a 3 unit fixed partial denture.



Figure 14.4 MODL complex amalgam tooth (mandibular first molar).

demand greater care in smaller teeth. Carelessness may increase the risk of pulpal irritation or exposure.

IV. **Interim restoration**
Complex amalgam restorations are sometimes indicated as interim restorations for teeth that require elaborate occlusal alterations, ranging from vertical dimension changes to correcting occlusal plane discrepancies. When cost of indirect restorations is a major factor for the patient, the complex direct amalgam restoration may be an appropriate treatment option, provided that adequate resistance and retention forms are included (Fig. 14.4).[2,3] For some older patients and/or those who are debilitated, complex amalgam restoration may be the treatment preferred over the more expensive and time-consuming indirect restoration.

Contraindications

The complex amalgam restoration may be contraindicated:

I. if the tooth cannot be restored properly with direct restoration because of anatomic or functional considerations (or both);
II. if the area to be restored has esthetic importance for the patient.

Advantages

Conservation of Tooth Structure

The preparation for a complex amalgam restoration is usually more conservative than the preparation for an indirect restoration.

Appointment Time

The complex restoration may be completed in one appointment. An indirect restoration generally requires at least two appointments unless it is done using a chairside computer-aided design/computer-assisted manufacturing (CAD/CAM) system.

Resistance and Retention Forms

Amalgam restorations with cusp coverage significantly increase the fracture resistance of weakened teeth compared with amalgam restorations without cusp coverage.[4] Resistance and retention forms may be significantly increased by the use of slots and pins (discussed in subsequent sections).

Reduced Cost

The complex amalgam restoration may be utilized to reinforce and stabilize compromised posterior teeth at a much reduced cost to the patient. It may serve as a definitive final restoration or an intermediate-term restoration with a long-term goal of indirect, more costly restoration and protection of the tooth.

Disadvantages

Tooth Anatomy

Proper contours and occlusal contacts and anatomy are sometimes difficult to achieve with large, complex restorations.

Resistance Form

Resistance form is more difficult to develop with a complex amalgam as compared to the preparation of a tooth for a cusp-covering onlay (skirting axial line angles of the tooth [Chapter 23]) or a full crown. The complex amalgam restoration does not protect the tooth from fracture as effectively as a full-coverage indirect restoration.

Tooth Preparation for Complex Amalgam Restorations

In this section, various types of complex amalgam preparation procedures are discussed before the restoration technique is presented. Also, the word 'vertical' is used to describe tooth preparation walls and other preparation aspects that are approximately parallel to the long axis of the tooth. The word 'horizontal' is used to describe the walls and other aspects that are approximately perpendicular to the long axis of the tooth. Every effort should be made to protect the pulp as early as possible when accomplishing the procedures associated with complex amalgam restorations.

Preparation for Cusp Coverage Complex Amalgams

Indications

- When a caries lesion is extensive with resultant loss of dentin necessary to support occlusally loaded areas of the tooth, reduction of one or more of the cusps followed by coverage with amalgam may be indicated.
- *When the facial or lingual extension exceeds two-thirds the distance from a primary fissure towards the cusp tip* (or when the faciolingual extension of the occlusal preparation exceeds two-thirds the distance between the facial and lingual cusp tips), reduction of the cusp for coverage with amalgam usually is required for the development of adequate resistance form (Fig. 14.5A). For cusps prone to fracture, coverage reduces the risk of fracture and extends the life of the restoration.[5,6] Complex amalgam restorations that cover one or more cusps have documented longevity of 72% after 15 years.[3,7]

Clinical Technique

- The operator uses the bur to reduce the cusp, following the mesiodistal inclines of the cusp; this results in a uniform reduction (Fig. 14.5B–E).
- If only one of two facial (or lingual) cusps is to be covered, the cusp reduction should extend slightly beyond the facial (or lingual) groove area, provide the correct amount of tooth structure removal and meet the adjacent, unreduced cusp to create a 90-degree cavosurface margin. This approach results in adequate thickness and edge strength of the amalgam (Fig. 14.5F and G).
- Cusp reduction and coverage reduce the amount of vertical preparation wall height and increase the need for the use of secondary retention features (Fig. 14.6). An increased retention form may be provided by proximal box retention grooves, but may require the use of slots and/or pins (as described in subsequent sections). If indicated, cusp reduction and coverage increase the resistance form of the tooth.[3,8]
- Reduction of the unsupported cusp should be accomplished during the initial tooth preparation because it improves access and visibility for subsequent steps. If the cusp to be reduced is located at the correct occlusal height before preparation, depth cuts should be made on the remaining occlusal surface of each cusp to be covered, using the side of a carbide fissure bur or a suitable diamond instrument (Fig. 14.5B).
- The *depth cuts should be a minimum of 2 mm for functional cusps and 1.5 mm for nonfunctional cusps*.[6] These dimensions provide adequate strength for amalgam by preventing flexure of the restorations during loading. To reduce the cusp, the dentist orients the bur parallel to the cuspal incline (lingual incline for facial cusp reductions and facial inclines for lingual cusp reduction) and makes several depth cuts in the cusp (to a depth of 1.5 or 2 mm as indicated). The depth cuts provide guides for the correct amount of cusp reduction. Without depth cuts, after the beginning reduction of the cusp, the operator may no longer know how much more reduction is necessary. When using the restoration to correct an occlusal relationship, if the unreduced cusp height is located at less than the correct occlusal height, the depth cuts may be less. Likewise, if the unreduced cusp height is located

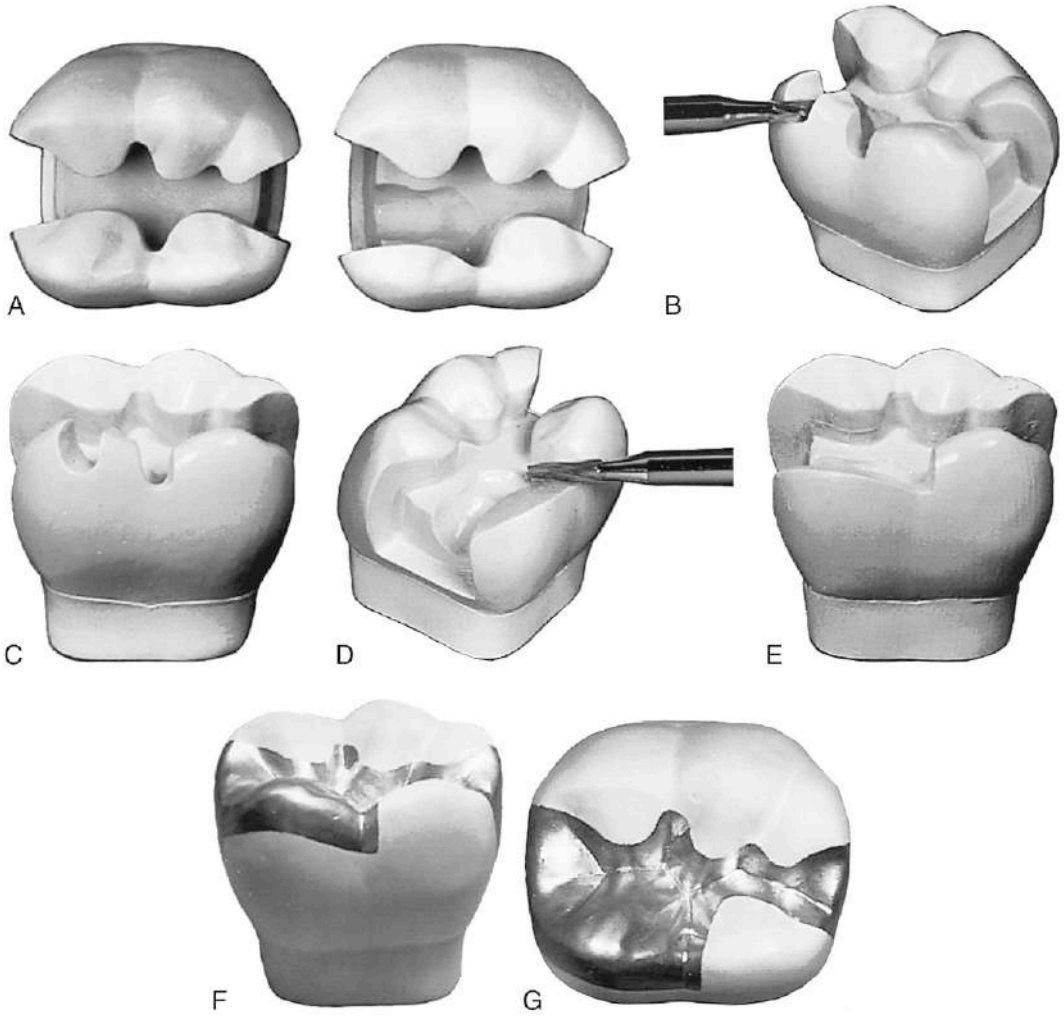


Figure 14.5 Covering the cusp with amalgam. (A) Comparison of the mesial aspects of normally extended (left) and extensive (right) mesioocclusodistal tooth preparation. The resistance form of the mesiolingual cusp with extensive preparation is compromised and cusp coverage with amalgam is indicated. (B) Preparing depth cuts. (C) Depth cuts prepared. (D) Reducing the cusp. (E) Cusp reduced. (F, G) Final restoration.

higher than the correct occlusal height, the depth cuts may be deeper.

- The goal is to ensure that the final restoration has restored cusps with a minimal thickness of *2 mm of amalgam for functional cusps* and *1.5 mm of amalgam for nonfunctional cusps* (Fig. 14.5C) while developing an appropriate occlusal relationship. Using the depth cuts as a guide, the reduction is completed to provide for a uniform reduction of tooth structure (Fig. 14.5D). The occlusal contour of the reduced cusp should be similar to the normal contour of the unreduced cusp.
- Any sharp internal corners of the tooth preparation formed at the junction of prepared surfaces should be rounded to reduce stress concentration in the amalgam and improve its resistance to fracture from occlusal forces (Fig. 14.5E).

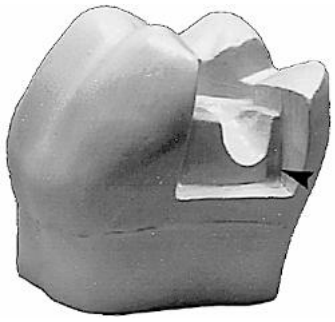


Figure 14.6 A retention groove is associated with the remaining vertical wall of a cusp requiring reduction.

Mandibular First Premolar

The lingual cusp of the mandibular first premolar may need to be reduced for coverage with amalgam if the lingual margin of the occlusal step extends more than two-thirds the distance from the central fissure to the cuspal eminence (Fig. 14.7). It is acceptable when restoring diminutive nonfunctional cusps, such as the lingual cusp



Figure 14.7 Mandibular first premolar with the lingual cusp reduced for coverage with amalgam.

of a mandibular first premolar, to reduce the cusp only 0.5–1 mm and restore the cusp to achieve an amalgam thickness of 1.5 mm. This procedure conserves more of the lingual wall of the isthmus for added retention form.

Maxillary First Molar

The procedure for reducing the distolingual cusp of a maxillary first molar for coverage is illustrated in Fig. 14.8. Extending the facial or lingual wall of a proximal box to include the entire cusp is accomplished (if necessary) to include weak or carious tooth structure or existing restorative material.

Mandibular First Molar

Reducing the distal cusp is an alternative to extending the entire distofacial wall when the occlusal margin crosses the cuspal eminence (Fig. 14.9). The cusp reduction should result in a butt joint between the tooth structure and amalgam.

Tooth Preparation for Slot-Retained Amalgam Restorations

Definition

A slot is a horizontal retention groove in dentin.

Indication

When loss of vertical coronal height is approximately 2–4 mm, use of slots, in horizontal areas where no vertical walls remain, may be indicated to gain secondary retention form for the complex amalgam restoration (Figs. 14.10–14.12).

Clinical Technique

- Slots are placed in the gingival floor of a preparation with a No. 330 bur (Fig. 14.10B, inset).
- In general, slots
  - should be about 1 mm wide and 1 mm deep;
  - should be placed in the line-angle areas of the tooth;
  - should be 2–4 mm in length (depending on the distance between remaining vertical walls);
  - should be positioned 0.5–1 mm inside the DEJ; and
  - One slot per missing axial line angle should be used.

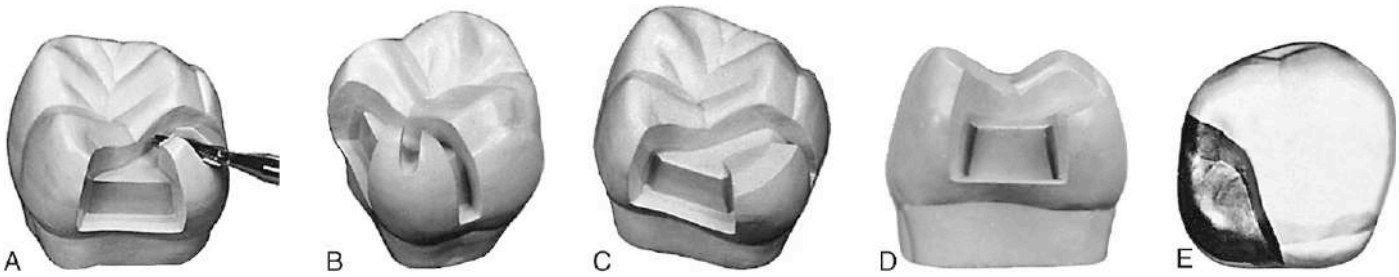


Figure 14.8 Reduction of the distolingual cusp of the maxillary molar. (A) Cutting a depth gauge groove with the side of the bur. (B) Completed depth gauge groove. (C, D) Completed cusp reduction. (E) Distoocclusolingual complex amalgam.

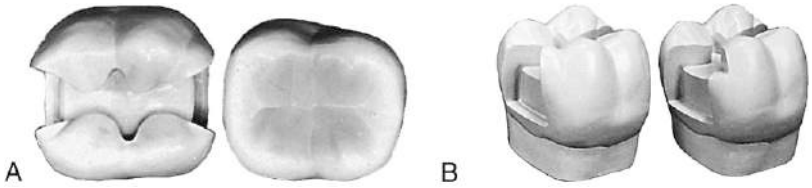


Figure 14.9 Mandibular first molar. (A) Reduction of the distal cusp is indicated when the occlusal margin crosses the cuspal eminence. (B) Distofacial view of the distal cusp shown in *A* before reduction for coverage (left); the distal cusp after reduction (right). A reduction of 2 mm is necessary to provide for minimal 2-mm thickness of amalgam.

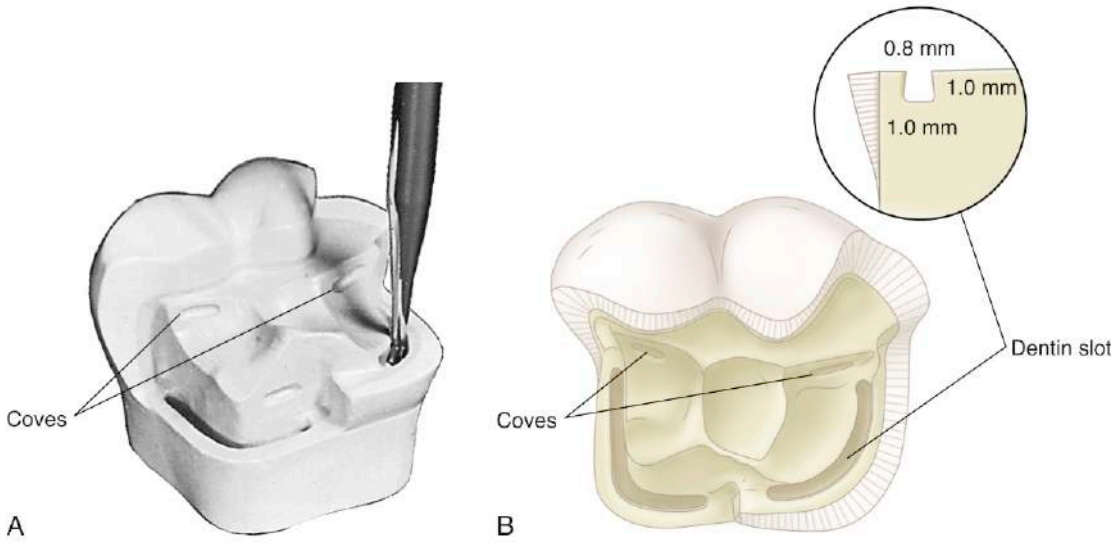


Figure 14.10 Slots. (A, B) With a No. 330 bur, dentinal slots are prepared approximately 1 mm deep and 0.5–1 mm inside the DEJ.



Figure 14.11 Coves prepared in dentin with No. ¼ bur, where appropriate.



Figure 14.12 Vertical grooves prepared in dentin with a No. ¼ round or No. 169L bur, where appropriate.

- Increasing the width and depth of the slot does not increase the retentive strength of the amalgam restoration.[9] Shorter slots provide as much resistance to horizontal force as do longer slots.[10] Preparations with shorter slots fail less frequently than do preparations with longer slots.[10]
- Slot design should be such that it is independently retentive as a singular feature (the internal walls of the slot should be convergent) (Fig. 14.10B, inset). In addition, the walls of the slot should work together with other vertical walls to create an overall convergence of the preparation so as to enhance retention of the restoration.
- In addition to any slot placement, retention grooves and coves may be utilized in association with remaining vertical walls (Figs. 14.13 and 14.14). Coves and retention grooves should be prepared when possible. Coves are prepared in a horizontal plane, and grooves are prepared in a vertical plane.

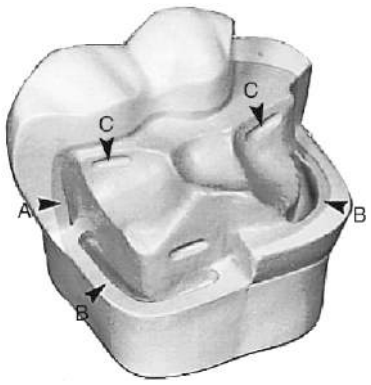


Figure 14.13 Groove (A), slots (B) and coves (C).

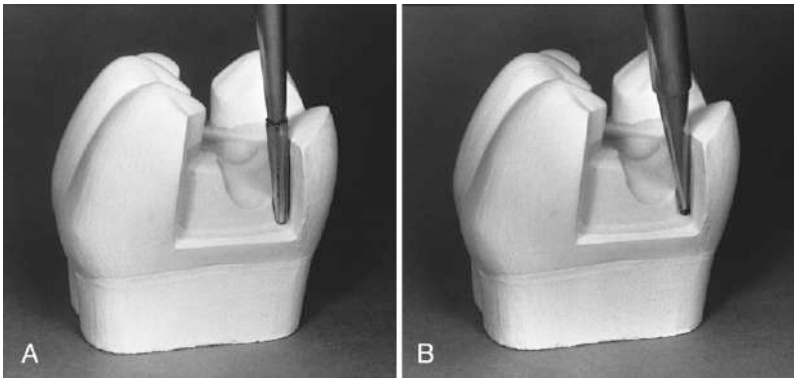


Figure 14.14 Placement of retention grooves. (A) Position of No. 169L bur to prepare the retention groove. (B) Groove prepared with No. ¼ bur.

Tooth Preparation for Pin-Retained Amalgam Restorations

Definition

A *pin-retained amalgam* (also referred to as a 'pin-amalgam') *is defined as any restoration requiring the placement of one or more pins in dentin to provide adequate resistance and retention forms.*

Indications

- When severe carious destruction or cusp fracture has resulted in the *loss of 4 mm or more of vertical coronal height*, consideration should be given to the use of a parapulpal retention pin to regain some of the lost vertical height.
- Pins are used whenever *adequate resistance and retention forms* are not able to be established with slots, grooves or undercuts only.[11] Pins placed into prepared pinholes (also referred to as *pin channels*) are used to provide auxiliary resistance and retention forms.
- The pin-retained amalgam is an important *adjunct* in the restoration of teeth with extensive caries lesions or fractures.[12]

Advantages

- Amalgam restorations including pins have significantly greater retention compared with restorations using boxes only or restorations relying solely on bonding systems.[13]

Disadvantages

- Caution is indicated when using pins. Preparing pinholes and placing pins may create *craze lines* or fractures and internal stresses in dentin.[14–16] Such craze lines and internal stress areas may have little or no clinical significance, but they may be important when minimal dentin is present.
- Preparation steps for pin retention have *increased risk of penetrating into the pulp* or perforating the external tooth surface.
- The use of pins *decreases the tensile and horizontal strength* of pin-retained amalgam restorations.[17,18]

Types of Parapulpal Retention Pins

There are three types of pins for pin-retained amalgam restorations (Fig. 14.15).

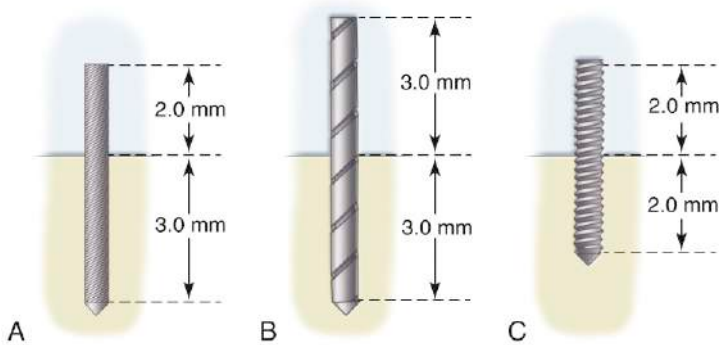


Figure 14.15 Three types of pins. (A) Cemented. (B) Friction locked. (C) Self-threading.

Cemented pins

I. In 1958, Markley described a technique for restoring teeth with amalgam and cemented pins using threaded (or serrated) stainless steel pins.
II. They are cemented into pinholes prepared *0.025–0.05 mm larger than the diameter of the pin*.
III. The cementing medium may be any standard dental luting agent.

Friction locked pins

I. In 1966, Goldstein described a technique for the friction-locked pin.
II. The diameter of the prepared pinhole is *0.025 mm smaller than the diameter of the pin*.
III. The pins are tapped into place, retained by the resiliency of the dentin.
IV. They are 2–3 times more retentive than cemented pins.

Clinical Note

Friction-locked and cemented pins, although still available, are rarely used.

Self-threading pins

- The *most frequently* used type of parapulpal pin is the self-threading pin.
- The pin-retained amalgam restoration using self-threading pins originally was described by Going in 1966.[19]
- The diameter of the prepared pinhole is 0.03–0.10 mm smaller than the diameter of the pin (Table 14.1). The threads engage dentin as the pin is inserted, thus retaining the pin. The elasticity (resiliency) of dentin permits insertion of a threaded pin into a hole of smaller diameter.[20]
- Although the threads of self-threading pins do not engage dentin for their entire width, self-threading pins are the most retentive of the three types of pins (Fig. 14.16), being 3–6 times more retentive than cemented pins.[21–23]

Vertical and horizontal stresses, potentially resulting in craze lines (cracks), are generated in dentin when a self-threading pin is inserted. Craze lines in dentin are related to the size of the pin. The insertion of 0.787-mm self-threading pins produces more dentinal craze lines than does the insertion of 0.533-mm self-threading pins.[24] Pulpal stress is maximal when the self-threading pin is inserted perpendicular to the pulp.[25]

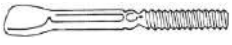
The depth of the pinhole varies from 1.3 to 2 mm, depending on the diameter of the pin used.[26] Recommended pinhole depth is generally accepted to be **2 mm**.

The **Thread Mate System (TMS)** (Coltène/Whaledent Inc., Cuyahoga, OH) is the most widely used self-threading pin system because of its

I. versatility,
II. wide range of pin sizes,
III. colour-coding system, and
IV. greater retentiveness.[27,28]

TMS pins are available in gold-plated stainless steel or in titanium. Titanium alloy pins (Max Restorative Pins, Coltène/Whaledent Inc.) are also available.

Table 14.1 The Thread Mate System (TMS) pins

Name	Illustration (not to scale)	Colour code	Pin diameter (inches/mm)*	Drill diameter (inches/mm)*	Total pin length (mm)	Pin length extending from dentin (mm)
Regular (standard)		Gold	0.031/0.78	0.027/0.68	7.1	5.1
Regular (self-shearing)		Gold	0.031/0.78	0.027/0.68	8.2	3.2
Regular (two-in-one)		Gold	0.031/0.78	0.027/0.68	9.5	2.8
Minim (standard)		Silver	0.024/0.61	0.021/0.53	6.7	4.7
Minim (two-in-one)		Silver	0.024/0.61	0.021/0.53	9.5	2.8
Minikin (self-shearing)		Red	0.019/0.48	0.017/0.43	7.1	1.5
Minuta (self-shearing)		Pink	0.015/0.38	0.0135/0.34	6.2	1

*1 mm = 0.03937 inch.

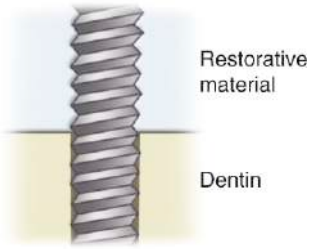


Figure 14.16 The complete width of the threads of self-threading pins does not engage dentin.

Factors Affecting Retention of the Pin in Dentin and Amalgam

Type

With regard to the retentiveness of the pin in dentin, the *self-threading pin is the most retentive*, the friction-locked pin is intermediate and the cemented pin is the least retentive.[21]

Surface characteristics

The number and depth of the elevations (serrations or threads) on the pin influence the retention of the pin in the amalgam restoration. The shape of the self-threading pin results in the greatest retention of amalgam.

Orientation, number and diameter

- Placing pins in a *nonparallel manner* increases their ability to retain the restoration; however, bending pins to improve their retention in amalgam is not advised as the bends may interfere with adequate condensation of amalgam around the pin and decrease amalgam retention. In addition, bending also may weaken the pin and risk fracture of the adjacent dentin.
- Pins should be bent only to provide for an adequate amount of amalgam (approximately 1 mm) between the pin and the external surface of the finished restoration (on its lateral surface). Only the manufacturer's specific bending tool should be used to bend a pin, not other hand instruments (Section, Pin Insertion).
- In general, increasing the *number of pins* increases their retention in dentin and amalgam. However, the potential benefit of increasing the number of pins must be compared with the potential problems. As the number of pins increases,
  - crazing of dentin and the potential for fracture increase;
  - amount of available dentin between the pins decreases; and
  - strength of the amalgam restoration decreases.[29,30]
- Also, as the *diameter of the pin* increases, retention in dentin and amalgam generally increases.
- As the number, depth and diameter of pins increase, the danger of perforating into the pulp or the external tooth surface increases. Numerous long pins may also severely compromise the condensation of amalgam and its adaptation to the pins. A pin technique (i.e. a balance between pin orientation, size and number) that permits optimal retention with minimal danger to the remaining tooth structure should be used.[31]

Extension into dentin and amalgam

Self-threading pin extension into dentin and amalgam should be approximately *1.5–2 mm* to preserve the strength of dentin and amalgam.[21] Extension greater than this is unnecessary for pin retention and is contraindicated.

Pin Placement Factors

Pin size

Two determining factors for selecting the appropriate-sized pin are

I. amount of dentin available to safely receive the pin, and
II. amount of retention desired.

Four sizes of TMS pins are available (Fig. 14.17), each with a corresponding colour-coded drill (Table 14.1). Familiarity with drill sizes and their corresponding colours is necessary to ensure that a proper-sized pinhole is prepared for the desired pin. It is difficult to specify a particular size of pin that is always appropriate for a particular tooth.

I. **Minikin (0.48 mm):** In the TMS system, the *pins of choice for severely involved posterior teeth* are the Minikin (0.48 mm) and, occasionally, the Minim (0.61 mm). The Minikin pins usually are selected to reduce the risk of dentin crazing, pulpal penetration and root surface perforation.
II. **Minim (0.61 mm):** The Minim pins are usually used as a backup in case the pinhole for the Minikin is overprepared or the pin threads strip dentin during placement resulting in a lack of Minikin pin retention in the dentin.
III. **Minuta (0.38 mm):** The Minuta pin is approximately half as retentive as the Minikin and one-third as retentive as the Minim pin.[27,28] The Minuta is usually too small to provide adequate retention in posterior teeth.
IV. **Regular (0.78 mm):** Larger diameter pins have the greatest retention.[32] However, the Regular or largest diameter pin is *rarely used* because a significant amount of crazing in the tooth (dentin and enamel) may be created during its insertion.[24,33] Of the four types of pins, the Regular pin is associated with the highest incidence of dentinal crack communication with the pulp chamber.[16]

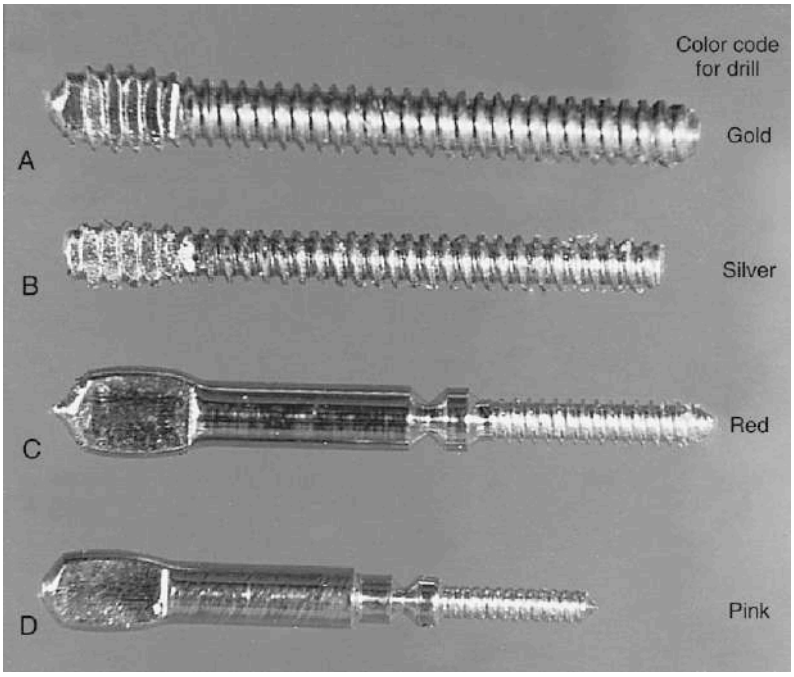


Figure 14.17 Four sizes of the Thread Mate System (TMS) pins. (A) Regular (0.78 mm). (B) Minim (0.61 mm). (C) Minikin (0.48 mm). (D) Minuta (0.38 mm).

Number of pins

Factors to be considered when deciding how many pins are required include

I. amount of missing tooth structure;
II. amount of dentin available to safely receive the pins;
III. amount of retention required; and
IV. size of the pins.

As a general rule, *one pin per missing axial line angle* should be used. The fewest pins possible should be used to achieve the desired retention for a given restoration. Although the retention of the restoration increases as the number of pins increases, an excessive number of pins may fracture the tooth and significantly weaken the amalgam restoration.

Pinhole location

Factors that aid in determining the pinhole locations include

I. knowledge of normal pulp anatomy and external tooth contours;
II. current radiograph of the tooth;
III. periodontal probe; and
IV. patient's age.

Although the radiograph is only a two-dimensional image of the tooth, it may give an indication of the position of the pulp chamber and the contour of the mesial and distal surfaces of the tooth. Consideration must also be given to the placement of pins in areas where the greatest bulk of amalgam will occur so as to compensate (through the reinforcing tendency of the amalgam restoration) for the weakening effect of the pins on the tooth structure.[34] Areas of occlusal contacts on the restoration must be anticipated because a pin oriented vertically and positioned directly below an occlusal contact point weakens amalgam significantly.[35] *Occlusal clearance should be sufficient to provide 2 mm of amalgam over the pin.*[36,37]

Several attempts have been made to identify the ideal location of the pinhole.[15,24,38,39] The following principles of pin placement are recommended:

- In the cervical third of molars and premolars (where most pins are located), pinholes should be located near the *line angles of the tooth* except as described later.[31,40]
- The pinhole should be positioned *no closer than 0.5–1 mm to the DEJ or no closer than 1–1.5 mm to the external surface of the tooth,* whichever distance is greater (Fig. 14.18). Before the final decision is made about the

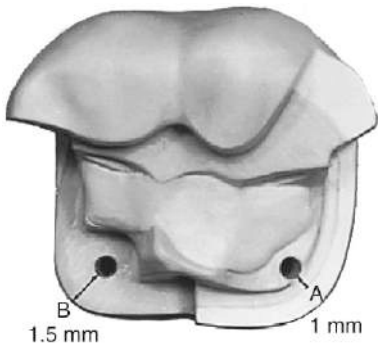


Figure 14.18 Pinhole position. (A) Position relative to the DEJ. (B) Position relative to external tooth surface.

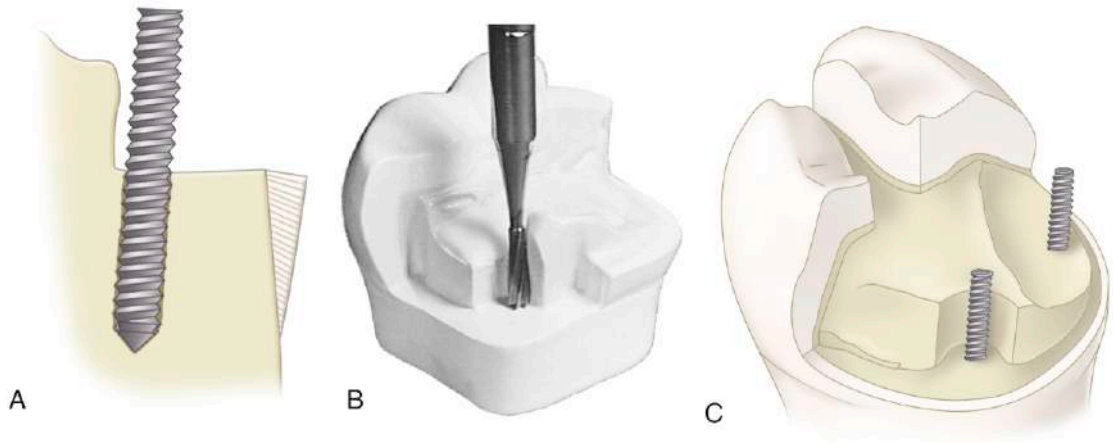


Figure 14.19 (A) Pin placed too close to the vertical wall such that adequate condensation of amalgam is jeopardized. (B, C) Recessed area prepared in the vertical wall of the mandibular molar with a No. 245 bur to provide adequate space for amalgam condensation around the pin.

location of the pinhole, the operator should probe the gingival crevice carefully to determine if any abnormal contours exist that would predispose the tooth to an external perforation. The pinhole should be parallel to the adjacent external surface of the tooth.

- The position of a pinhole must not result in the pin being so close to a vertical wall of tooth structure that condensation of amalgam against the pin or wall is jeopardized (Fig. 14.19A). It may be necessary to first prepare a recess in the vertical wall with the No. 245 bur to permit proper pinhole preparation and to provide a minimum *0.5-mm clearance around the circumference of the pin* for adequate condensation of amalgam (Fig. 14.19B and C).[41] If necessary, after a pin is inappropriately placed, the operator should provide clearance around the pin to provide sufficient space for the smallest condenser nib to ensure that amalgam is condensed adequately around the pin. A No. 169L bur may be used, taking care not to damage or weaken the pin.
- Pinholes should be prepared on a *flat surface* that is perpendicular to the proposed direction of the pinhole. This will help prevent the drill tip from slipping or 'crawling', and will allow a depth-limiting drill (discussed later) to prepare the pinhole as deeply as intended (Fig. 14.20).
- Whenever three or more pinholes are placed, they should be located at *different vertical levels on the tooth*, if possible; this reduces stresses resulting from pin placement in the same horizontal plane of the tooth. Spacing between pins, or the interpin distance, must be considered when two or more pinholes are prepared. The optimal interpin distance depends on the size of pin to be used. The minimal interpin distance is 3 mm for the Minikin pin and 5 mm for the Minim pin.[40] Maximal interpin distance results in lower levels of stress in dentin.[42]
- Several posterior teeth have anatomic features that may preclude safe pinhole placement. Fluted and furcal areas should be avoided.[40] External perforation may result from pinhole placement
  - over the prominent mesial concavity of the maxillary first premolar;
  - at the midlingual and midfacial bifurcations of the mandibular first and second molars; and
  - at the midfacial, midmesial and middistal furcations of the maxillary first and second molars.
- Pulpal penetration may result from pin placement
  - at the mesiofacial corner of the maxillary first molar and the mandibular first molar;
  - at posterior teeth that are rotated in the arch;
  - at teeth that are abnormally tilted in the arch; and

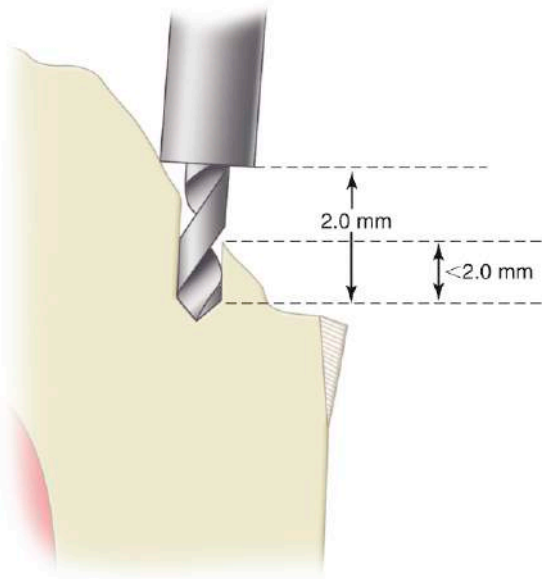


Figure 14.20 Use of a depth-limiting drill to prepare a pinhole in the surface that is not perpendicular to the direction of the pinhole results in a pinhole of inadequate depth.

- when possible, the location of pinholes on the distal surface of mandibular molars and lingual surface of maxillary molars should be avoided.

Obtaining the proper direction for preparing a pinhole in these locations is difficult because of the abrupt flaring of the roots just apical to the CEJ (Fig. 14.21). If the pinhole is placed parallel to the external surface of the tooth crown in these areas, penetration into the pulp is likely.[40] For mandibular second molars that are severely tilted mesially, care must be exercised to orient the drill properly to prevent external perforation on the mesial surface and pulpal penetration on the distal surface (Fig. 14.22). Because of limited interarch space, it is sometimes difficult to orient the twist drill (Section, Pinhole Preparation Technique) correctly when placing pinholes at the distofacial or distolingual line angles of the mandibular second and third molars (Fig. 14.23).

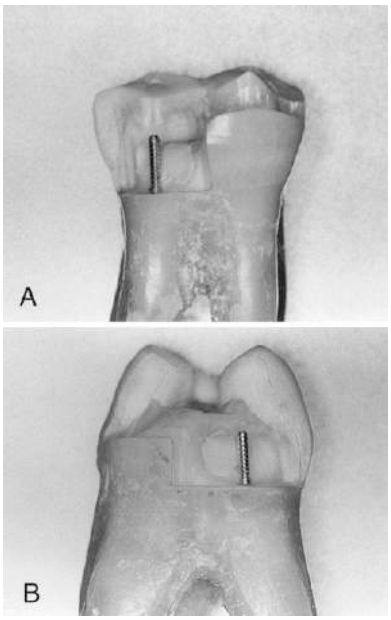


Figure 14.21 Distal flaring of the mandibular molar (A) and palatal root flaring of the maxillary molar (B). Root angulation should be considered before pinhole placement.

Pinhole Preparation Technique

When the pinhole locations have been determined, a No. ¼ round bur is first used to prepare a pilot hole (dimple) approximately one-half the diameter of the bur at each location (Fig. 14.24). The purpose of this pilot hole is to permit more accurate placement of the twist drill and to prevent the twist drill from moving laterally (i.e. 'crawling') when it has begun to rotate.

The KODEX drill (a twist drill provided in the TMS system) should be used for preparing pinholes (Fig. 14.25). The aluminium shank of this drill, which acts as a heat absorber, is colour coded to allow ready matching with the appropriate pin size (Table 14.2; see also Table 14.1). The drill shanks for the Minuta and Minikin pins are tapered to provide a built-in 'wobble' when placed in a latch-type contra-angle hand-piece. This wobble allows the drill to be 'free floating' and to align itself as the pinhole is prepared. This feature minimizes the potential for dentinal crazing or the breakage of small drills.

Because the optimal depth of the pinhole into the dentin is 2 mm (only 1.5 mm for the Minikin pin), a depth-limiting drill should be used to prepare the hole (Fig. 14.25). This type of drill is able to prepare the pinhole to the correct depth only when used on a flat surface that is perpendicular to the drill. When the location for starting a pinhole is not perpendicular to the desired pinhole direction, the location area should be flattened or the standard twist drill should be used (Fig. 14.25). The standard twist drill has blades that are 4–5 mm in length, which would allow preparation of a pinhole with an effective depth. Creation of a *flat area* and use of the *depth-limiting drill* provide for a more predictable outcome and are recommended.

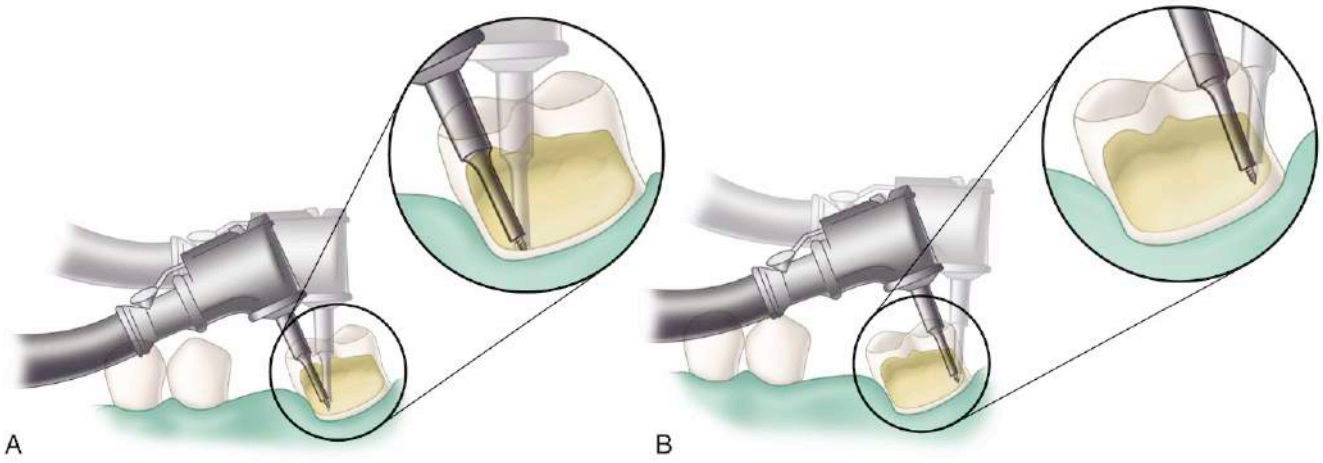


Figure 14.22 Care must be exercised when preparing pinholes in mesially tilted molars to prevent external perforation on mesial surface (A) and pulpal penetration on the distal surface (B).

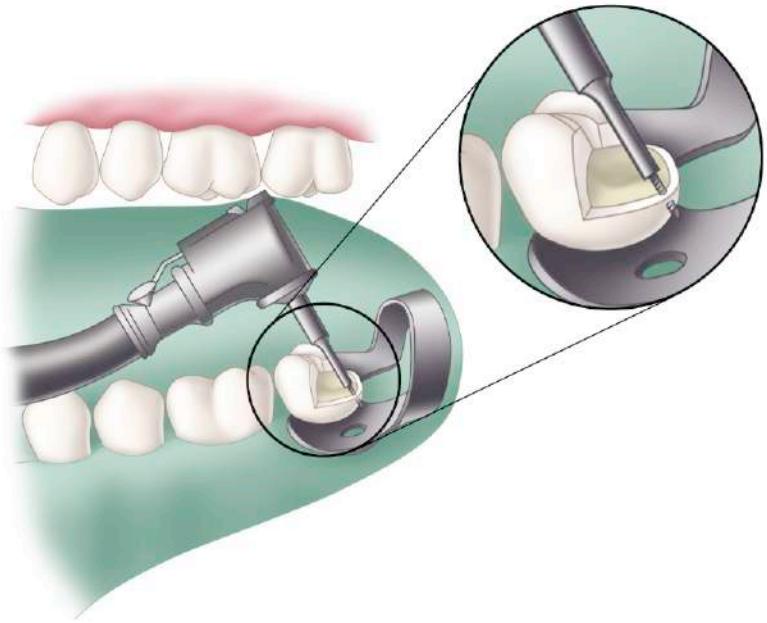


Figure 14.23 When placing pinholes in molars and interarch space is limited, care must be exercised to prevent external perforation on distal surface.



Figure 14.24 Pilot hole (dimple) prepared with a No. ¼ bur.

The technique of creating a properly aligned pinhole is as follows:

I. A pilot hole (dimple) is placed in the horizontal dentin surface to receive the pin.
II. The correct size twist drill (based on the pin size) is placed in a latch-type contra-angle hand-piece.
III. The twist drill is placed in the gingival crevice beside the location for the pinhole and positioned such that it lies flat against the external surface of the tooth.
IV. Without changing the angulation obtained from the gingival crevice position, the hand-piece is moved occlusally and the drill is placed in the previously prepared pilot hole (Fig. 14.26A).
V. The drill is then viewed from a 90-degree angle to the previous viewing position to ascertain that the drill also is angled correctly in this plane (Fig. 14.26B).
VI. With the drill tip in its proper position and with the hand-piece rotating at very low speed (300–500 rpm), pressure is applied to the drill until the depth-limiting portion of the drill is reached (Fig. 14.26C and D).
VII. The pinhole is prepared in one or two movements *without allowing the drill to stop rotating*. While still rotating, the drill is immediately removed from the pinhole.

Incorrect angulation of the drill may result in pulpal exposure or external perforation. If the proximity of an adjacent tooth interferes with placement of the drill into the gingival crevice, a flat, thin-bladed hand instrument is placed into the crevice and against the external surface of the tooth to indicate the proper angulation for the drill.[43] Using more than one or two movements, tilting the hand-piece during the drilling procedure, or allowing the drill to rotate more than briefly at the bottom of the pinhole will result in a pinhole that is too large. The twist drill should never stop rotating (from insertion to removal from the pinhole) so as to limit the potential for binding and/or breakage while creating the pinhole.

Dull twist drills used to prepare pinholes may cause increased frictional heat and cracks in the dentin. The signal for discarding the drill is the need for increased pressure on the hand-piece.[44] Using a drill when its self-limiting shank shoulder has become rounded is contraindicated (Fig. 14.27).

Pin Design

Several designs are available for each of the five sizes of TMS self-threading pins: standard, two-in-one, Link Series, Link Plus and self-shearing (Fig. 14.28).

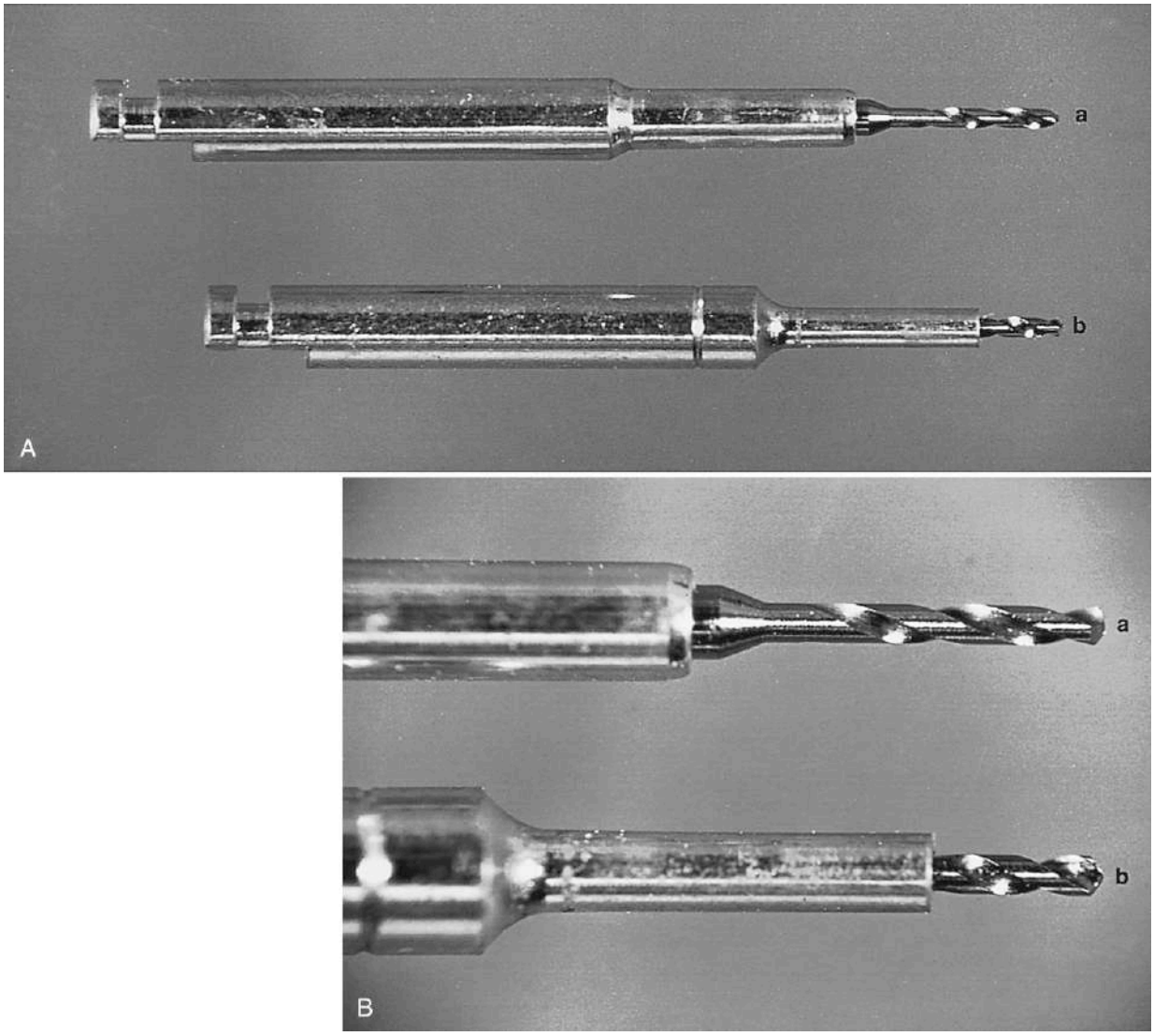


Figure 14.25 (A) Two types of KODEX twist drills: standard (*a*) and depth limiting (*b*). (B) Drills enlarged: standard (*a*) and depth limiting (*b*).

Table 14.2 The Thread Mate System (TMS) Link Series and Link Plus pins

Name	Illustration (not to scale)	Colour code	Pin diameter (inches/mm)*	Drill diameter (inches/mm)*	Pin length extending from sleeve (mm)	Pin length extending from dentin (mm)
Link Series						
Regular		Gold	0.031/0.78	0.027/0.68	5.5	3.2 (single shear)
Regular		Gold	0.031/0.78	0.027/0.68	7.8	2.6 (double shear)
Minim		Silver	0.024/0.61	0.021/0.53	5.4	3.2 (single shear)
Minim		Silver	0.024/0.61	0.021/0.53	7.6	2.6 (double shear)
Minikin		Red	0.019/0.48	0.017/0.43	6.9	1.5 (single shear)
Minuta		Pink	0.015/0.38	0.0135/0.34	6.3	1 (single shear)
Link Plus						
Minim		Silver	0.024/0.61	0.021/0.53	10.8	2.7 (double shear)

*1 mm = 0.03937 inch.

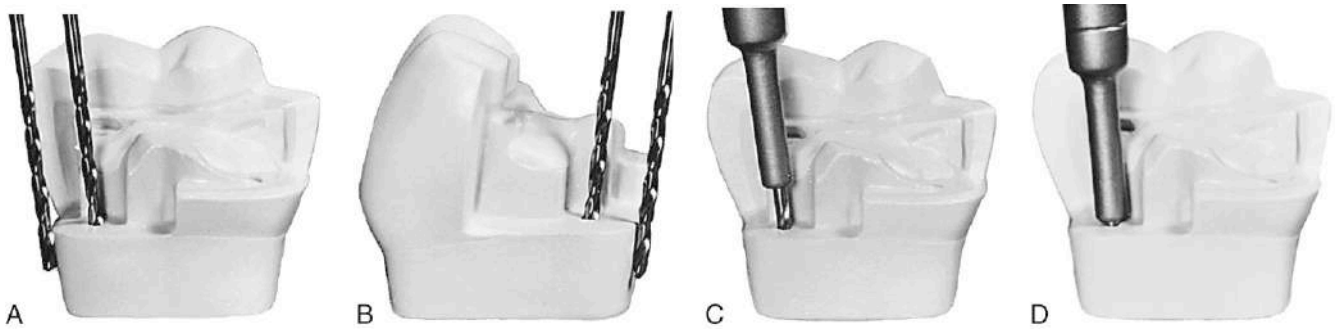


Figure 14.26 Determining the angulation for the twist drill. (A) Drill placed in the gingival crevice, positioned flat against the tooth and moved occlusally into position without changing the angulation obtained. (B) (A) repeated while viewing the drill from position 90 degrees left or right of that viewed in (A). (C, D) With twist drill at correct angulation, the pinhole is prepared in one or two thrusts until the depth-limiting portion of drill is reached and then removed while still rotating.

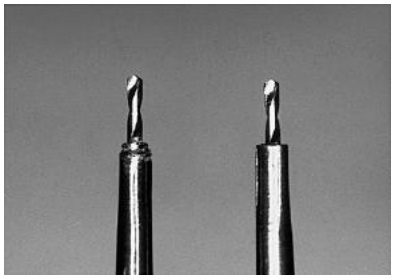


Figure 14.27 Minikin self-limiting drill with worn shank shoulder (left) compared with a new drill with an unworn shoulder (right).

Link Series

The Link Series pin is free floating in a plastic sleeve, which allows self-alignment as it is threaded into the pinhole (Fig. 14.29). When the pin reaches the bottom of the hole, the pin shears off at a machined narrow area, leaving a length of pin extending from dentin. The plastic sleeve is then discarded. The Minuta, Minikin, Minim and Regular pins are available in the Link Series. Link Series pins are recommended because of their versatility, self-aligning ability and retentiveness.[27]

Link Plus pins

Link Plus self-threading pins are also self-shearing and are available as single and two-in-one pins contained in colour-coded plastic sleeves (Fig. 14.30). This design has a sharper thread, a shoulder stop at 2 mm and a tapered

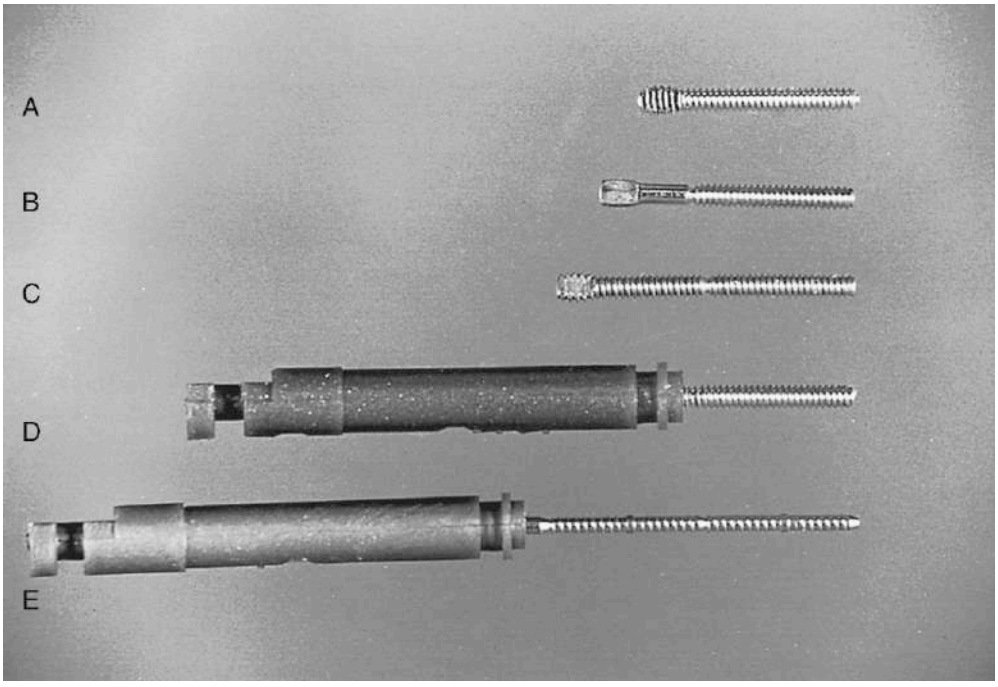


Figure 14.28 Five designs of the Thread Mate System (TMS) pins. (A) Standard. (B) Self-shearing. (C) Two-in-one. (D) Link Series. (E) Link Plus.



Figure 14.29 Cross-sectional view of Link Series pin.

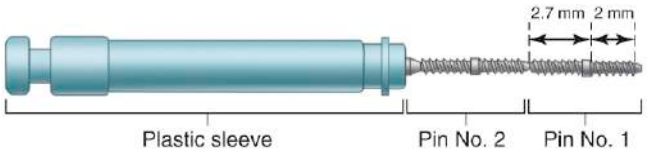


Figure 14.30 Link Plus pin.

tip to fit the bottom of the pinhole more readily as prepared by the twist drill. It also provides a 2.7-mm length of pin to extend out of dentin, which usually needs to be shortened with the use of a hand-piece bur. Theoretically, these innovations should reduce the stress created in the surrounding dentin as the pin is inserted and reduce the apical stress at the bottom of the pinhole.[44,45]

Self-shearing pins

The self-shearing pin has a total length that varies according to the diameter of the pin (Table 14.1). It also consists of a flattened head to engage the hand wrench or the appropriate hand-piece chuck for threading into the pinhole. When the pin approaches the bottom of the pinhole and begins to bind, the head of the pin shears off leaving a length of pin extending from dentin.

Two-in-one pins

The two-in-one pin is actually two pins in one, with each one being shorter than the standard pin. The two-in-one pin is approximately 9.5 mm in length and has a flattened head to aid in its insertion. When the pin reaches the bottom of the pinhole, it shears approximately in half, leaving a length of pin extending from dentin with the other half remaining in the hand wrench or the hand-piece chuck. This second pin may be positioned in another pinhole and threaded to place in the same manner as the standard pin. The designs available with each size of pin are shown in Tables 14.1 and 14.2.

Selection of a particular pin design is influenced by the size of the pin being used, the amount of interarch space available and operator preference. With minimal interarch space, the two-in-one design may be undesirable because of its length. It has been reported that 93% of Link Series and Link Plus two-in-one pins extended to the optimal depth of 2 mm.[46–49] Even so, both the two-in-one pin and the self-shearing pin may occasionally fail to reach the bottom of the pinhole.

Pin Insertion

The TMS utilizes *hand wrenches* to place the self-threading pins (Figs. 14.31A–C and 14.32A). Link Series and Link Plus pins are contained in colour-coded plastic sleeves that fit a latch-type contra-angle hand-piece or a specially

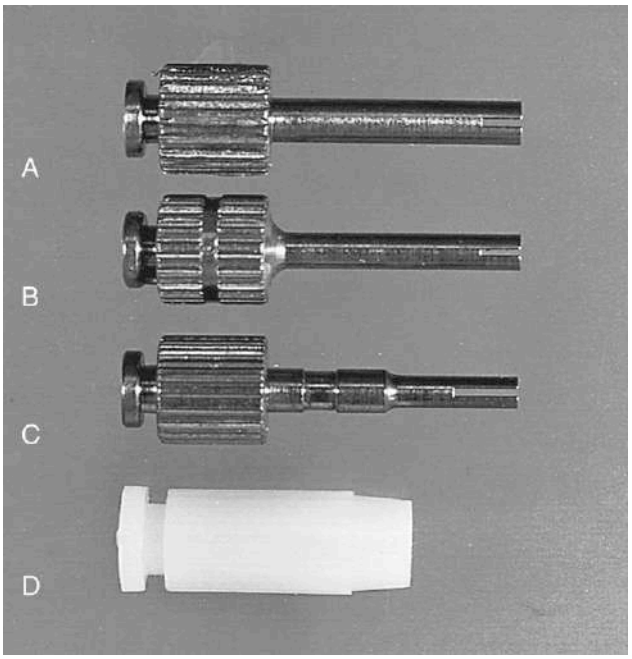


Figure 14.31 Hand wrenches for the Thread Mate System (TMS) pins. (A) Regular and Minikin. (B) Minim. (C) Minuta. (D) Link Series and Link Plus.



Figure 14.32 (A) Use of a hand wrench to place a pin. (B) Threading the pin to the bottom of the pinhole and reversing the wrench one quarter to one half turn. (C) Evaluating the length of the pin extending from dentin.

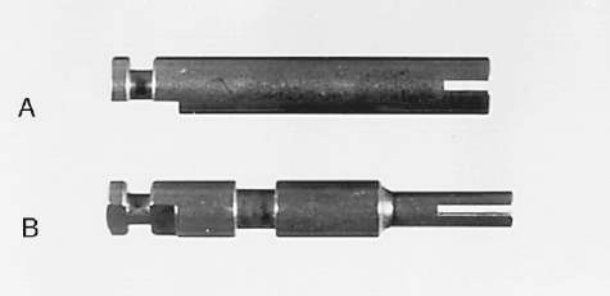


Figure 14.33 Hand-piece chucks for the Thread Mate System (TMS) regular self-shearing and Minikin pins (A) and TMS Minuta pins (B).

designed plastic hand wrench (Fig. 14.31D). In addition, hand-piece chucks are available (Fig. 14.33).

- The latch-type hand-piece is recommended for the insertion of the Link Series and the Link Plus pins.
- The hand wrench is recommended for the insertion of standard pins.

When using the latch-type hand-piece, a Link Series or a Link Plus pin is inserted into the hand-piece and positioned over the pinhole. The hand-piece is activated at low speed until the plastic sleeve shears from the pin. The pin sleeve is then discarded. For low-speed hand-pieces with a low gear, the low gear should be used. Using the low gear increases the torque and increases the tactile sense of the operator. It also reduces the risk of stripping the threads in dentin during pin insertion.

A standard pin is placed in the appropriate hand wrench (Fig. 14.32A) and slowly threaded clockwise into the pinhole until a definite resistance (indicating the pin has reached the bottom of the pin hole) is felt (Fig. 14.32B). The pin should then be rotated one-quarter to one-half turn counterclockwise to reduce the dentinal stress created by the end of the pin that is pressing on dentin at the bottom of the pinhole.[50,51] The hand wrench should be removed from the pin carefully.

The lengths of the pins are evaluated after placement (Fig. 14.32C). *Any length of pin greater than 2 mm should be removed.* A sharp No. ¼, ½, or No. 169L or 245 bur, at high speed and oriented perpendicular to the pin, may be used to remove the excess length (Fig. 14.34A). If oriented otherwise, the rotation of the bur may loosen the pin by rotating it counterclockwise and unthreading it out of the pinhole. After placement, the pin should be tight, immobile and not easily withdrawn. Every pin must be assessed with cotton pliers for stability immediately after placement.

The preparation is viewed from all directions with a mirror (particularly from the occlusal direction) to determine if any pins need to be bent so as to be positioned within the anticipated contour of the final restoration such that there will be adequate bulk of amalgam between the pin and the external surface of the final restoration (Fig. 14.34B and C).

When pins require bending, the *TMS bending tool* (Fig. 14.35 A) must be used. The bending tool should be placed on the pin where the pin is to be bent, and with firm controlled pressure, the bending tool should be rotated until the desired amount of bend is achieved (Fig. 14.35B–D). Use of the bending tool allows placement of the fulcrum at some point along the length of the exposed pin. Pins should only be bent if absolutely necessary, never simply to make them parallel or based on a notion that bending them will increase their relative retentiveness.

Failure of Pin-Retained Restorations

The failure of pin-retained restorations might occur at any of five different locations (Fig. 14.36). Failure may occur

I. within the restoration (restoration fracture);
II. at the interface between the pin and the restorative material (pin–restoration separation);
III. within the pin (pin fracture);
IV. at the interface between the pin and dentin; and
V. within dentin (dentin fracture).

Failure is more likely to occur at the pin–dentin interface than at the pin–restoration interface. The operator must keep these areas of potential failure in mind at all times and apply necessary principles to reduce failure risk.



Figure 14.34 (A) Use of sharp No. ¼ bur held perpendicular to the pin to shorten the pin. (B, C) Evaluating the preparation to determine the need for bending the pins.

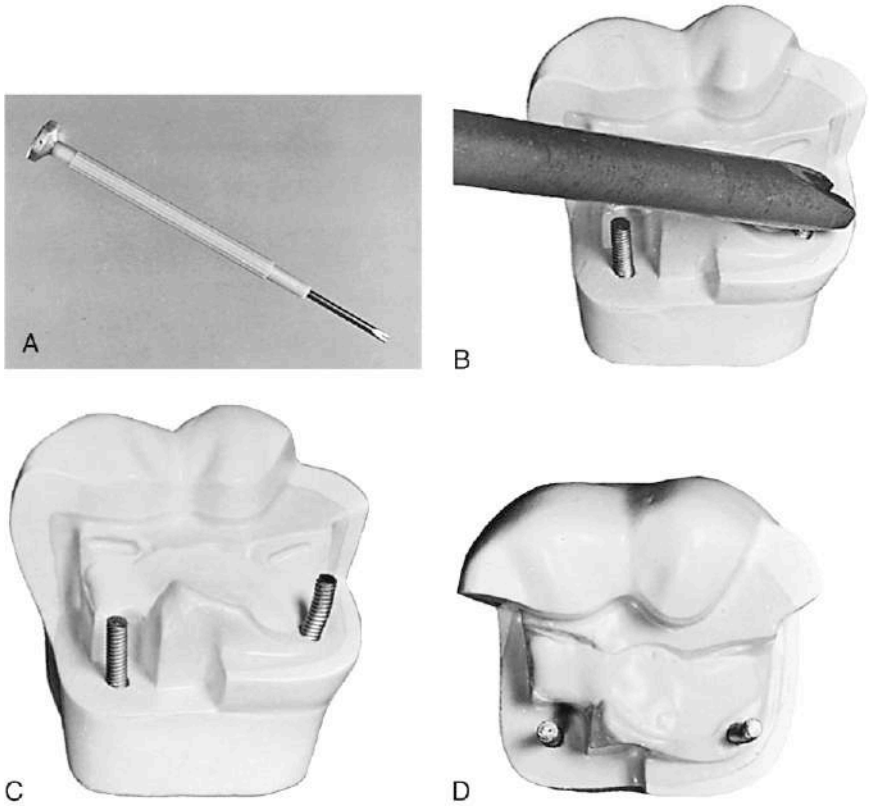


Figure 14.35 (A) The Thread Mate System (TMS) bending tool. (B) Use of the bending tool to bend the pin. (C, D) The pin is bent to a position that provides adequate bulk of amalgam between the pin and the external surface of the final restoration.

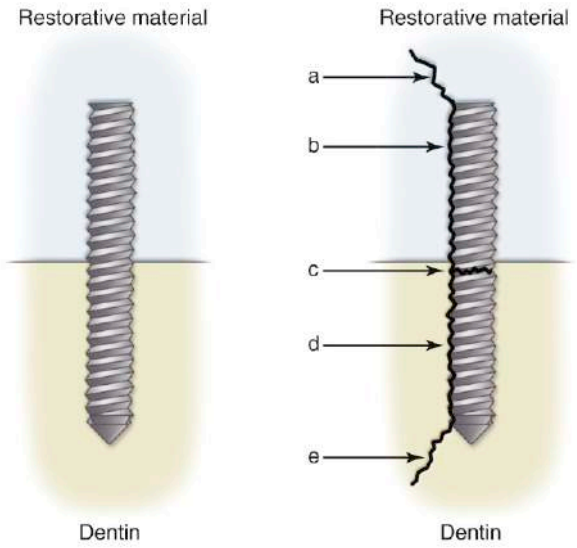


Figure 14.36 Five possible locations of failure of pin-retained restorations: fracture of restorative material (*a*), separation of pin from restorative material (*b*), fracture of pin (*c*), separation of pin from dentin (*d*), and fracture of dentin (*e*).

Considerations for the Use of Slots or Pins

Pin retention is used more frequently in preparations with few or no vertical walls. Slots are particularly indicated in short clinical crowns and when cusp reduction has been limited to 2 or 3 mm.[52]

More tooth structure is removed in slot preparation as compared with pin placement; however, slots are ***less likely*** to (i) create microfractures in dentin, (ii) perforate the tooth surface or (iii) penetrate into the pulp. Slot placement does not elicit an inflammatory response, whereas medium-sized self-threading pins may elicit an inflammatory response if placed within 0.5 mm of the pulp.[38]

Clinical Note

The retention potential of slots and pins is similar.[53–56]

Tooth Preparation for Amalgam Foundations

An *amalgam foundation* may be used as part of the initial restoration of a severely involved and serves to replace lost tooth structure necessary to provide retention and

resistance forms for the final indirect restoration (e.g. crown or bridge retainer).

The retention of the foundation material should not be compromised by tooth reduction during the final preparation for the indirect restoration. The foundation should also provide the resistance form against forces that otherwise might fracture the remaining tooth structure.

In contrast to a conventional complex amalgam restoration, an amalgam foundation may not depend primarily on the remaining coronal tooth structure for support. Instead, it may rely mainly on secondary preparation retention features (proximal retention grooves, coves, slots, pins).

A temporary or caries-control restoration may serve as a foundation, but only if the retention and resistance forms of the restoration are appropriate. A temporary or *caries-control restoration* is used to restore a tooth when

I. definitive treatment is uncertain;
II. several teeth require immediate attention for control of caries lesions; and
III. a tooth's prognosis is questionable.

A temporary or control restoration might depend only on the remaining coronal tooth structure for support, however, using few auxiliary retention features. When preparing a tooth for either a foundation or a temporary restoration, remaining unsupported enamel may be left (except at the gingival aspect) to aid in forming a matrix for amalgam condensation. In this case, the remaining unsupported enamel will subsequently be removed when the preparation for the indirect restoration is accomplished. Occasionally, when providing a temporary or control restoration, sufficient retention and resistance forms are included in the preparation to meet the requirements of a foundation.

As a rule, *foundations are placed in anticipation of full-coverage indirect restorations*. Amalgam and composite resin are the direct restorative materials used for foundations. RMGIs may be used to block out minor undercut areas of the preparation for an indirect restoration that do not draw. Amalgam may be preferred by some clinicians because it is easy to use and it is strong. Although slots and threaded pins may be used for secondary retention in vital teeth, prefabricated posts and cast post/cores are used to provide additional retention for the foundation material in endodontically treated teeth receiving foundations. Prefabricated posts and cast post/cores are generally used on anterior teeth or single-canal premolars that have little or no remaining coronal tooth structure. Note, however, that there must be adequate remaining natural tooth structure to allow creation of a *ferrule* so as to achieve long-term clinical success. The pulp chamber and entrance to the root canals of endodontically treated molar teeth typically provide retention for foundations, and it is not necessary to use any form of intraradicular retention (i.e. post material extended further into the root canal space) (Fig. 14.37).

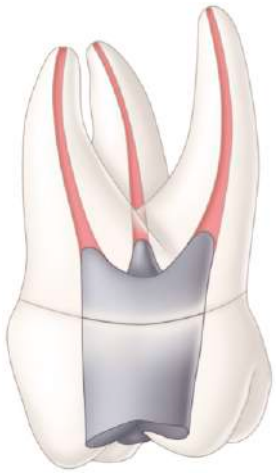


Figure 14.37 Pulp chamber retention with 2- to 4-mm extension of the foundation into the canal spaces.

Restorative Technique for Complex Amalgam Preparations

Desensitizer Placement

A dentin desensitizer is placed over the prepared tooth structure per manufacturer instructions. Excess moisture is removed without desiccating the dentin.

Matrix Placement

One of the most difficult steps in restoring a severely compromised posterior tooth is development of a satisfactory matrix to assist in formation of the anatomic shape of the restoration. Fulfilling the objectives of a matrix is complicated by possible gingival extensions, missing line angles and restoration of reduced cusps typical of complex tooth preparations.

Universal Matrix

The Tofflemire retainer and band may be used successfully for most amalgam restorations (Fig. 14.38). Use of the Tofflemire retainer requires sufficient tooth structure to retain the band after it is applied. Even when the Tofflemire retainer is placed appropriately an opening may remain next to an area of the preparation tooth structure. In this case, a closed system may be developed as illustrated in Fig. 14.39. A strip of matrix material that is long enough to extend from the mesial to the distal corners of the tooth is prepared. The strip must extend into these corners sufficiently that the band, when tight, holds the strip in position. Also, it must not extend into the proximal areas, or a ledge would result in the restoration contour (Fig. 14.39F). The Tofflemire retainer is loosened one-half turn, and the strip of matrix material is inserted next to the opening between the matrix band and the tooth. The

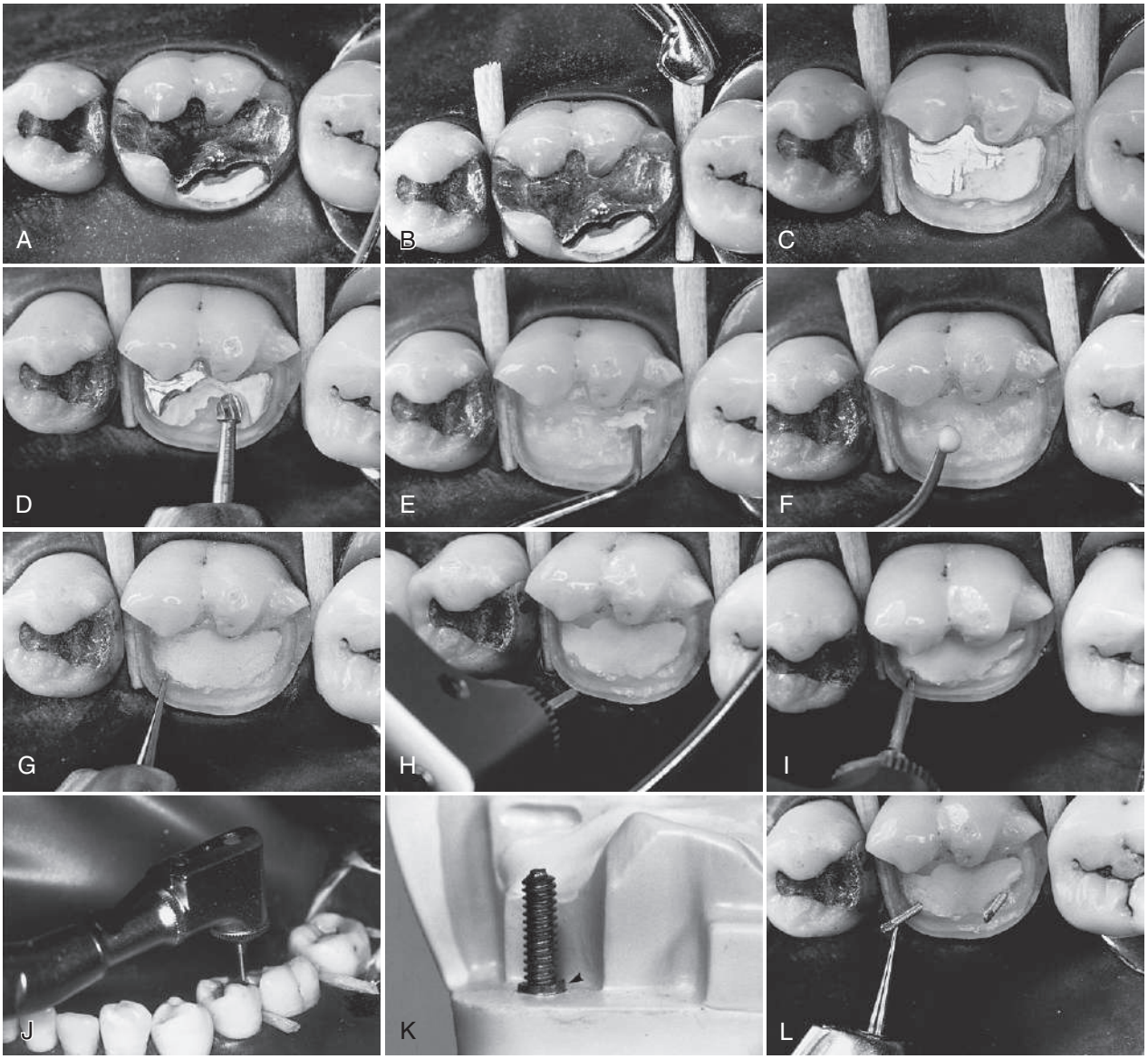


Figure 14.38 (A) Mandibular first molar with fractured distolingual cusp. (B) Insertion of wedges. (C) Initial tooth preparation. (D, E) Removal of any soft dentin and/or any remaining old restorative material as indicated. (F) Application of a liner and a base (if necessary). (G) Preparation of pilot holes. (H) Alignment of the twist drill with the external surface of the tooth. (I) Preparation of pinholes. (J) Insertion of Link pins with a slow-speed hand-piece. (K) Depth-limiting shoulder *(arrowhead)* of inserted Link Plus pin. (L) Use of a No. ¼ bur to shorten pins. (M) Bending pins (if necessary) with a bending tool. (N) Final tooth preparation. (O) Tofflemire retainer and matrix band applied to the prepared tooth. (P) Reflecting light to evaluate the proximal area of the matrix band. (Q) Preparation overfilled. (R) Restoration carved. (S) Reflecting light to evaluate the adequacy of the proximal contact and contour. (T) Restoration polished.

retainer is then tightened and the matrix is completed. Sometimes it is helpful to place a small amount of rigid material (hard-setting polyvinyl siloxane [PVS] or compound or light-cured flowable resin) between the strip and the open aspect of the band retainer to stabilize and support the strip (Fig. 14.39G and H).

When little tooth structure remains and deep gingival margins are present, the Tofflemire matrix may not function successfully, and the Automatrix system (DENTSPLY Caulk, Milford, DE) may be an alternative method (Fig. 14.40).

Automatrix

The Automatrix is a *retainerless matrix system* designed for any tooth regardless of its circumference and height. The Automatrix bands are supplied in three widths:

- 4.8 mm
- 6.35 mm
- 7.79 mm

The 4.8 mm, 6.5 mm and 7.79 mm bandwidths are available in 0.05 mm thickness only.

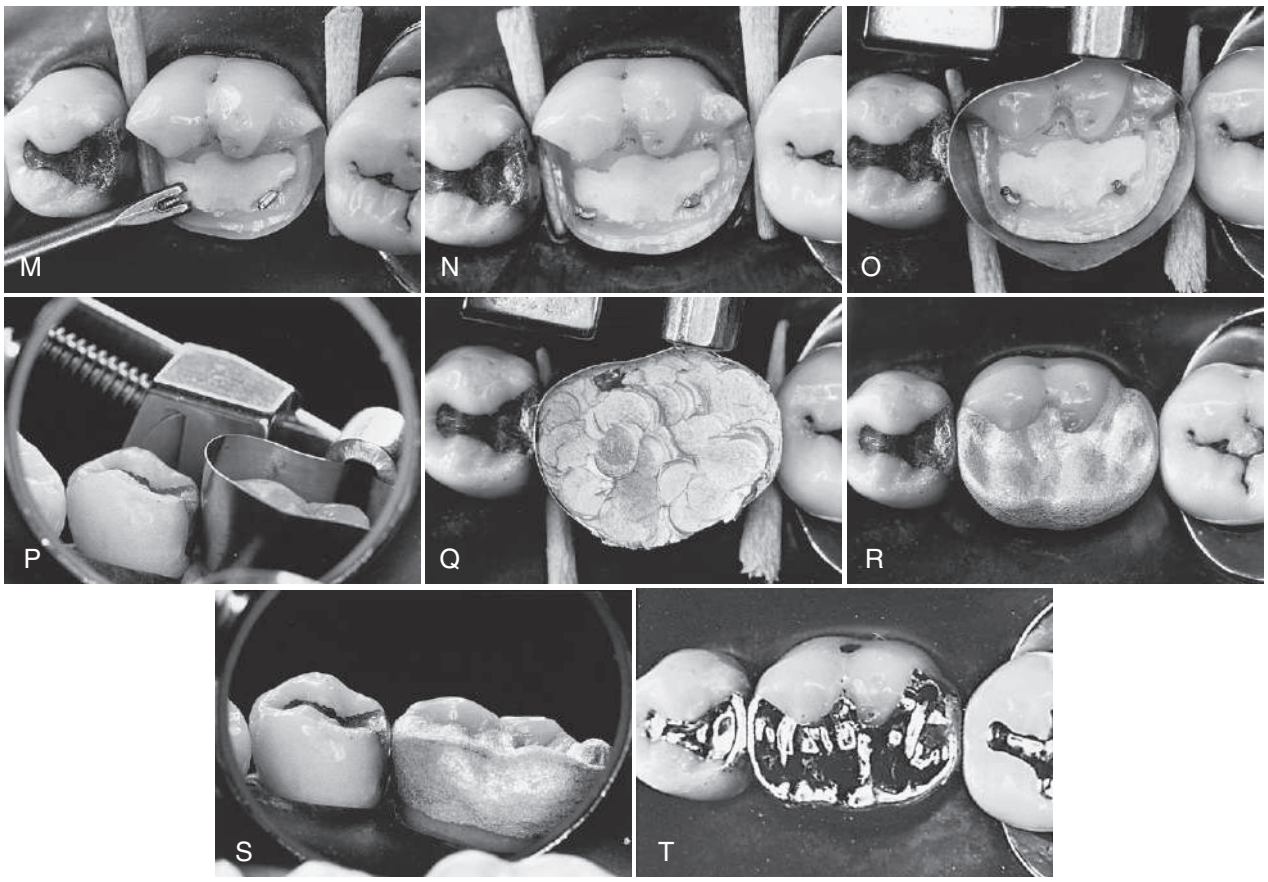


Figure 14.38 (*Cont.*)

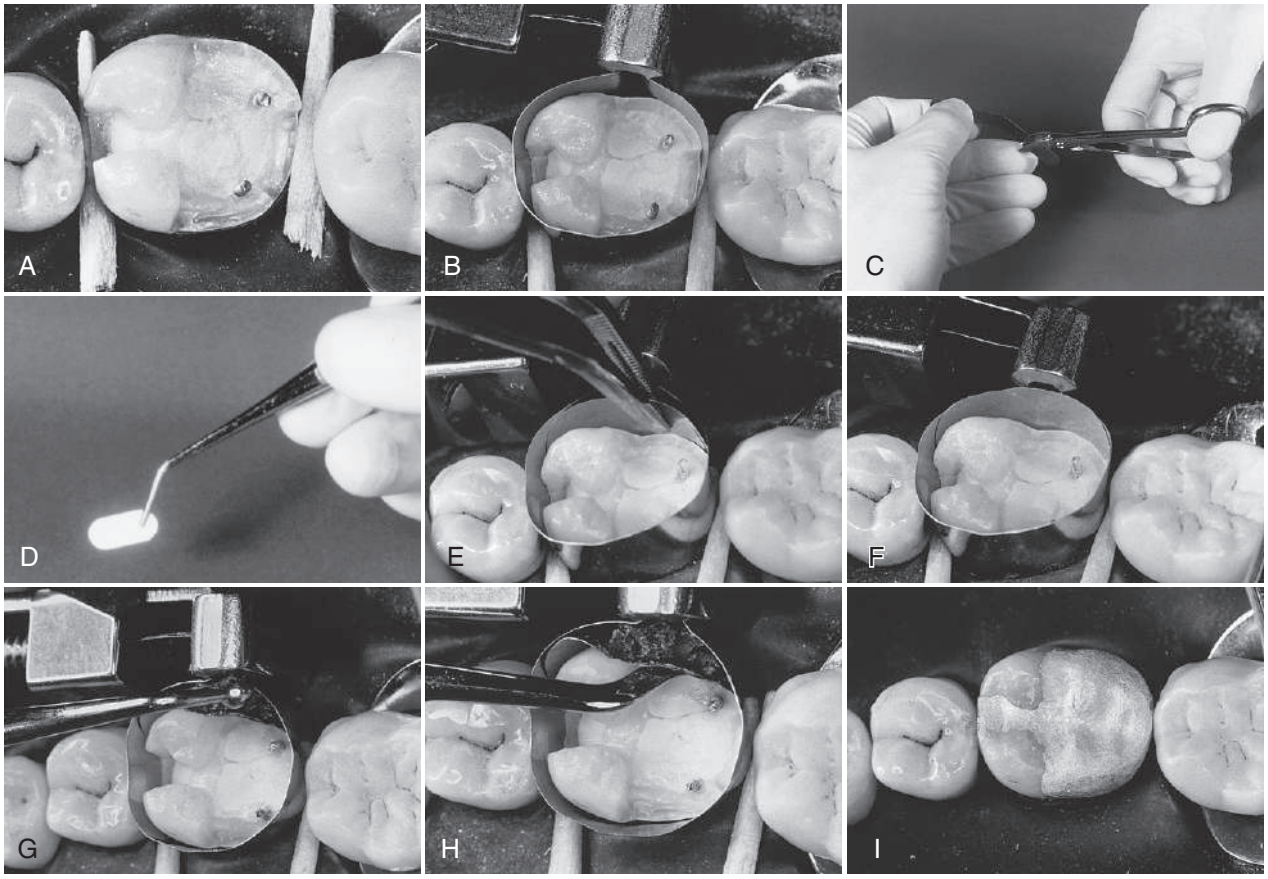


Figure 14.39 Technique for closing the open space of the Tofflemire matrix system. (A) Tooth preparation with wedges in place. (B) Open aspect of the matrix band next to the prepared tooth structure. (C, D) Cutting an appropriate length of the matrix material. (E) Insertion of a strip of the matrix material. (F) Closed matrix system. (G, H) Placement of the rigid supporting material between the strip and the matrix band, and contouring, if necessary. (I) Restoration carved.



Figure 14.40 (A) Automatrix retainerless matrix system. (B) Automatrix band. (C) Automate II tightening device. (D) Shielded snippers. (Part A: Courtesy Dentsply Caulk, Milford, DE.)

Advantages of this system include

- convenience;
- improved visibility because of absence of a retainer; and
- ability to place the autolock loop on the facial or lingual surface of the tooth.

Disadvantages of this system are that

- the band is flat and difficult to burnish and is sometimes unstable even when wedges are in place, and
- development of proper proximal contours and contacts may be difficult.

Use of the Automatrix system is illustrated in Fig. 14.41. Regardless of the type of matrix system used,

- the matrix must be stable during condensation of the amalgam;
- the matrix should extend beyond the gingival margins of the preparation enough to provide support for the matrix and to permit appropriate wedge stabilization; and
- the matrix should extend occlusally beyond the marginal ridge of the adjacent tooth by 1–2 mm.

Insertion, Contouring and Finishing of Amalgam

A *high-copper alloy* is strongly recommended for the complex amalgam restoration because of excellent clinical performance and high early compressive strengths.

I. *Spherical alloys* have a higher early strength than admixed alloys, and spherical alloys may be condensed more quickly and with less pressure so as to ensure good adaptation in slots and around the pins.
II. However, proximal contacts may be easier to achieve with *admixed alloys* because of their condensability.

Alloys that provide extended working time are necessary to allow adequate time for condensation, removal of the matrix band and final carving. Because complex amalgam

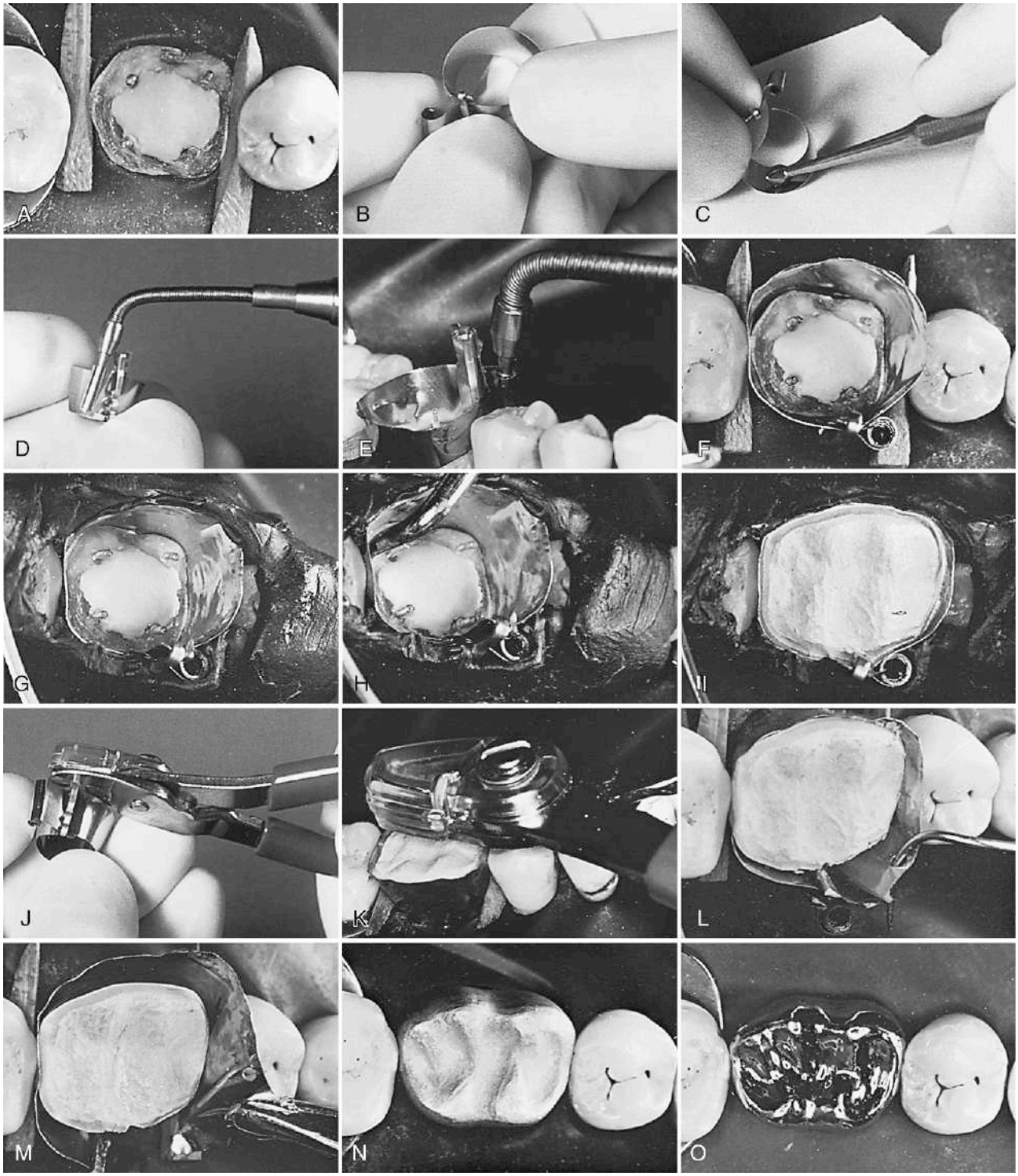


Figure 14.41 Application of Automatrix for developing a pin-retained amalgam crown on the mandibular first molar. (A) Tooth preparation with wedges in place. (B) Enlargement of the circumference of the band, if necessary. (C) Burnishing the band with an egg-shaped burnisher. (D–F) Placement of the band around the tooth, tightening with an Automate II tightening device and setting wedges firmly in place. (G) Application of the green compound. (H) Contouring of the band with the back of a warm Black spoon excavator so the softened compound will allow matrix deformation. (I) Overfilling the preparation and carving the occlusal aspect. (J, K) Use of shielded snippers to cut an autolock loop. (L) Separating the band with an explorer. (M) Removing the band in an oblique direction (facially with some occlusal vector). (N) Restoration carved. (O) Restoration polished.

restorations usually are quite large, a slow-set or medium-set amalgam may be selected to provide more time for the carving and adjustment of the restoration.

Regardless of the type of alloy chosen, care must be taken to condense the amalgam thoroughly in and around the retentive features of the preparation. If a mix of amalgam becomes dry or crumbly, a new mix is triturated immediately. Placement of a whole freshly mixed mass of amalgam (instead of using the amalgam carrier to place increments of amalgam) increases the rate of completely filling the defect. Special care must be taken when this technique is used to ensure the amalgam is well adapted in and around all retentive areas and at all margins.

With a complex (or any large) amalgam, *carving time* must be properly allocated. The time spent on carving occlusal anatomy (while the matrix is still in place) must be shortened to allow adequate time for carving the more inaccessible gingival margins and the proximal and axial contours (after the matrix is removed). The bulk of excess amalgam on the occlusal surface is removed and the initial anatomic contours are rapidly developed, especially the marginal ridge heights and cuspal inclines, with a *discoid carver* and/or *Hollenback carver*. The occlusal embrasures are defined by running the tine of an explorer against the internal aspect of the matrix band. Appropriate marginal ridge heights and embrasures reduce the potential of fracturing the marginal ridge when the matrix is removed.

Matrix removal is crucial when completing complex amalgam restorations, especially slot-retained restorations. If the matrix is removed prematurely, the newly placed restoration may fracture immediately adjacent to the areas where amalgam has been condensed into the slot(s). Any rigid material supporting the matrix is removed with an explorer or a Black spoon. Tofflemire matrices are removed first by loosening and removing the retainer, while the wedges are still in place. Leaving the wedges in place may help prevent fracturing the freshly condensed amalgam. It may be beneficial to place a fingertip on the occlusal surface of the restored tooth to stabilize the matrix while loosening and removing the retainer from the band. This is to reduce the risk of fracturing the freshly inserted amalgam while applying the force necessary to loosen the retainer. The matrix is removed by sliding each end of the band in an *oblique direction* (i.e. moving the band facially or lingually while simultaneously moving it in an occlusal direction). Moving the band obliquely towards the occlusal surface minimizes the possibility of fracturing the marginal ridge. Preferably the matrix band should be removed in the same direction as the wedge placement to prevent dislodging the wedges. Automatrix bands are removed by using the system's instruments and, after the band is open, by the same technique described for Tofflemire-retained matrices. Carving of the restoration is then continued (Figs. 14.38R and N).

Wedges are then removed, and the gingival excess on the proximal surface is removed with an amalgam knife or explorer. Facial and lingual contours are developed with a Hollenback carver, amalgam knife or explorer to complete the carving (Figs. 14.38R and S). Appropriately shaped rotary instruments are used to complete the occlusal contouring if amalgam has become so hard that the force needed to carve with hand instruments might fracture portions of the restoration.

Thin, unwaxed dental floss may be carefully passed through the proximal contacts one time to remove amalgam shavings and to smooth the proximal surface of amalgam. Amalgam excess and loose particles are removed from the gingival sulcus by moving the floss occlusogingivally and faciolingually. The patient should be cautioned not to chew with the restoration for 24 h. Fast-setting high-copper amalgam may be prepared for an indirect restoration within 30–45 min after insertion of the foundation. Further finishing and polishing of the complex amalgam may be accomplished, if desired, as early as 24 h after placement.

Summary

Complex amaglam preparations use retention grooves/coves in remaining vertical walls as well as slots, pins and customized matrix designs to effect successful restorations.

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